

Castalia

A four-channel electrochemical sensing SoC with a five-hart RISC-V back end

Maxx Seminario

University of Nebraska–Lincoln

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Where cancers are caught, and whether anyone looks

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Target
application

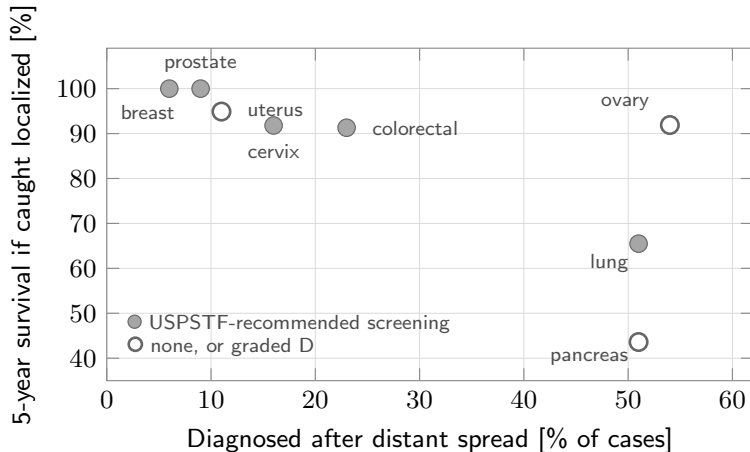
The chip

Analog front end

From currents to
a score

References

Backup



[1] SEER 21, 2016–2022; [2, 3] USPSTF

Why ovarian: late, lethal, and unscreened

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Target application

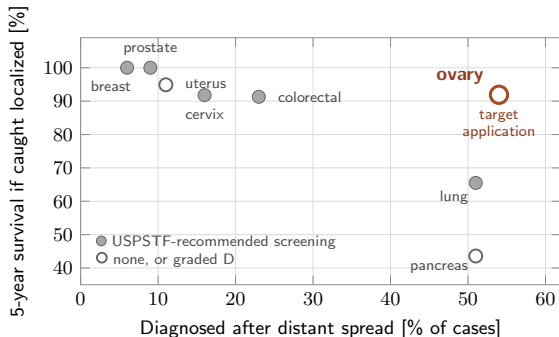
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Site	Distant at dx	Deaths/100 cases	Screening (USPSTF)
Ovary	54 %	59	D
Pancreas	51 %	78	D
Lung	51 %	54	B, LDCT
Colorectal	23 %	35	A
Cervix	16 %	31	A
Uterus	11 %	21	none
Prostate	9 %	11	C, PSA
Breast	6 %	13	B

USPSTF: A/B recommended; C offered selectively; D recommended *against*; none: no statement [7].

[1] SEER, 2026 estimates; [2, 3] USPSTF

Most lethal gynecologic cancer [1]: 92 % survivable if caught localized, yet 54 % present after spread, and screening is graded D [2]. UKCTOCS moved stage, not mortality [4].

One serum aliquot, four sites, one score

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Target application

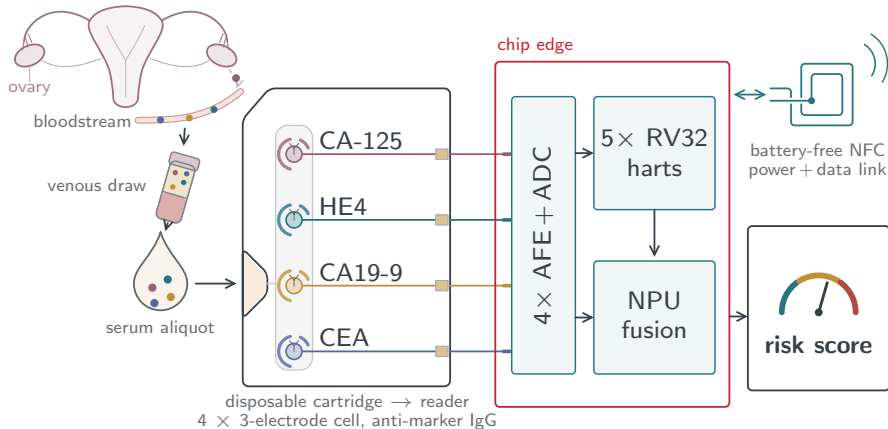
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Symptomatic-patient triage on a disposable four-site cartridge; four markers read at the same instant, fused on chip.

No single ovarian marker is a usable test [5]

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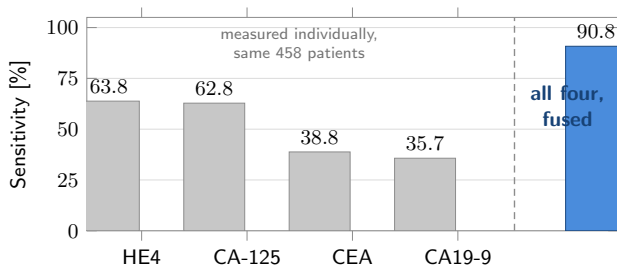
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Panel	Sensitivity
Best single (HE4)	63.8 %
Four fused	90.8 %
ROMA, FDA-cleared [6]	2 markers
This chip	4 channels

Clinical data (458 patients), not simulation. The decision is a learned combination, so the unit of integration is four concurrent channels plus the processor fusing them.

Four markers resolved in one differential-pulse sweep

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Target application

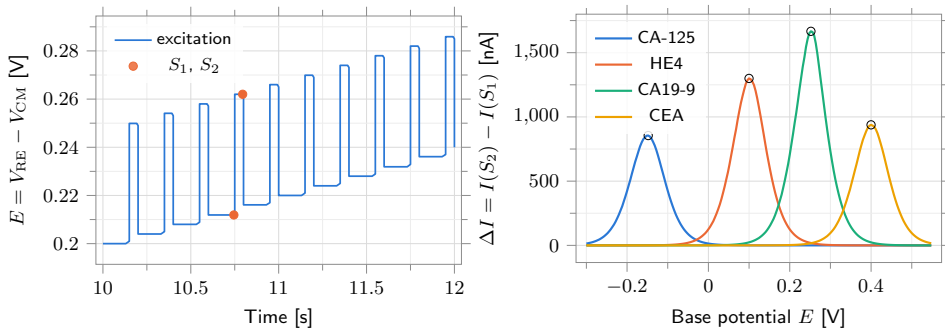
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■ schematic sim, pre-layout

Peak position identifies the reporter, peak height carries concentration. Apexes within 2 mV and 1 % of the assay targets; blank below 1 pA.

Every clinical cut-off lands inside the channel window

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Target
application

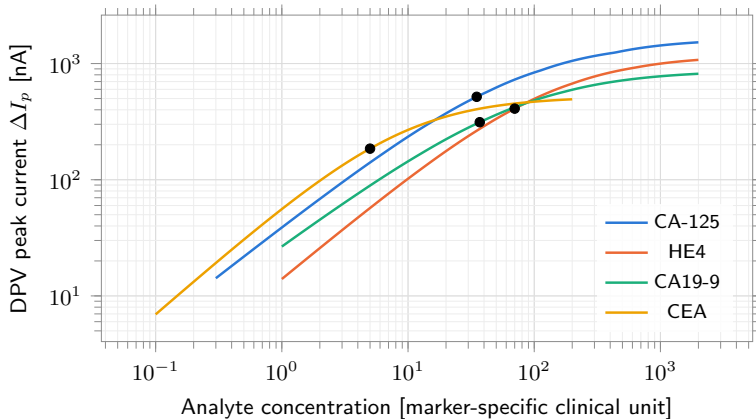
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■ schematic sim,

pre-layout

CA-125 35 U/mL, HE4 70 pmol/L, CA19-9 37 U/mL, CEA 5 ng/mL: all four on the steep part of the curve at one R_f setting.

Castalia at a glance

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Front currents to
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Digital

Harts	5 (1 orchestrator, 4 channel tiles)
ISA	rv32imac_zba_zbb_zbs_zbc
Clock	24 MHz rated, closed at 25 MHz
Measured CPI	1.260 aggregate, nine benchmarks
Private TCM	8 KiB per hart, 1-cycle
Shared RAM	80 KiB behind a round-robin arbiter
Synchronisation	16 hardware mutexes, LR/SC, AMOs
Interrupts	claim/complete, any vector to any hart
Accelerator	NPU: MLP, conv-1D, XNOR, GEMM
Power domains	4 MTCMOS, one per channel tile
Debug	JTAG, RISC-V Debug Module
Field boot	NFC ISO 14443A harvested boot

Analog and physical

Measurement sites	4 identical, simultaneous
Topology	bipolar grounded-WE potentiostat
Electrode pads	12 (CE/RE/WE $\times$ 4)
Converter	10-bit SAR per channel, extracted
Reference DAC	12-bit R-2R per channel
Feedback resistor	$18\text{ k}\Omega + 6\text{ k}\Omega \cdot N$, 320 codes
Full scale	$\pm 40.7\text{ }\mu\text{A}$ to $\pm 0.41\text{ }\mu\text{A}$
EIS	firmware mode of the same channels
Analog supply	2.5 V , $v_{\text{cm}} = 1.25\text{ V}$
Process	65 nm CMOS, 1P8M, HVT cells
Die / package	$3.0 \times 3.0\text{ mm}$ / LQFP-100
Digital power	10.72 mW post-route, SS, 25 MHz

Digital numbers from the hardened flow; analog numbers schematic-level, except the SAR, which is extracted.

Block diagram

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Target
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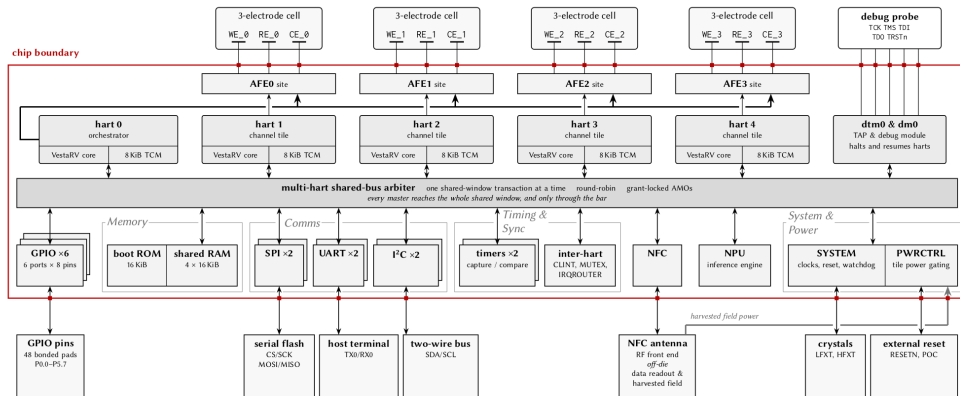
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Analogue front end

From currents to
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Four channel tiles own four AFE sites; every master reaches ROM, RAM, NPU and peripherals only through the arbiter.

Physical implementation

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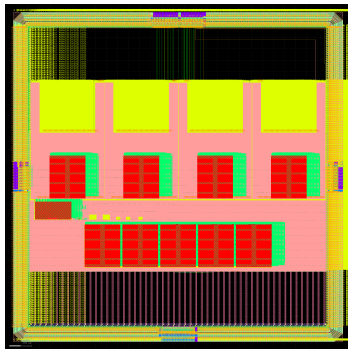
The chip

Analog front end

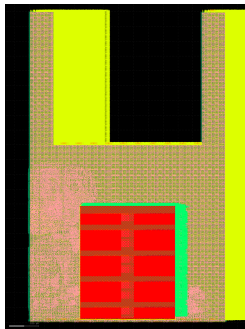
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chip top, 3.0 × 3.0 mm



one hardened channel tile

Process	65 nm, 1P8M, HVT
Core box	2690 × 2690 μm
Tile	660 × 880 μm
Tile logic	≈98 k NAND2-eq.
Tile SRAM	8 KiB
Package	LQFP-100, 14 × 14 mm

■ hardened flow, post-route

One tile hardened once, instantiated four times; the notch in each tile is the analog channel's keep-out.

Digital, as closed

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Whole chip, post-route

Internal	3.78 mW
Switching	2.33 mW
Leakage	4.61 mW
Total	10.72 mW

One channel tile

Total	1.761 mW
Gated	0

Measured CPI, timed kernels

Benchmark	Instr.	CPI
spmv	802 447	1.143
multiply	20 892	1.163
vvadd	2 408	1.374
median	4 015	1.398
qsort	123 205	1.398
rsort	146 548	1.412
dhrystone	200 523	1.504
towers	3 747	1.682
memcpy	11 017	1.726
Aggregate	1 314 802	1.260

Timing at 25 MHz

reg2reg WNS	+99 ps
Leakage share	43 %

■ hardened flow, post-route

SS 0.9 V 125 °C, statistical activity 0.2. Gating one tile removes 1.76 mW; the always-on band pays nothing.

Grounded-WE two-amplifier potentiostat

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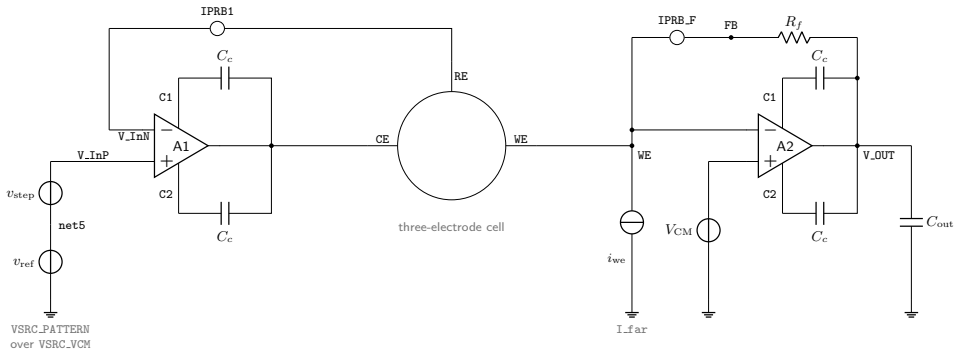
The chip

Analog front end

From currents to
a scope

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Backup



A1 drives CE with RE in feedback; A2 holds WE at v_{cm} through R_f . $v_{out} = v_{cm} - I_{we}R_f$, both polarities.

Bipolar accuracy through zero

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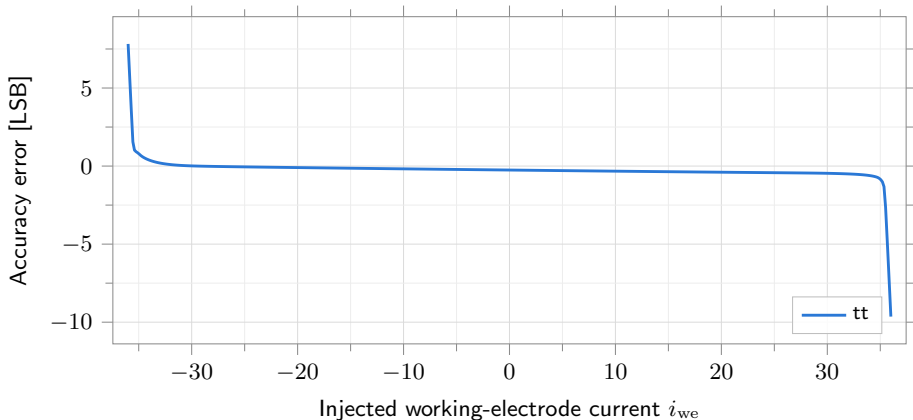
The chip

Analog front end

From currents to a scope

References

Backup



■ schematic sim, pre-layout

No kink at the crossing: class-AB output, no dead zone. This is the measurement rev-1's PMOS mirror could not make.

Cyclic voltammetry through the assembled channel

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application

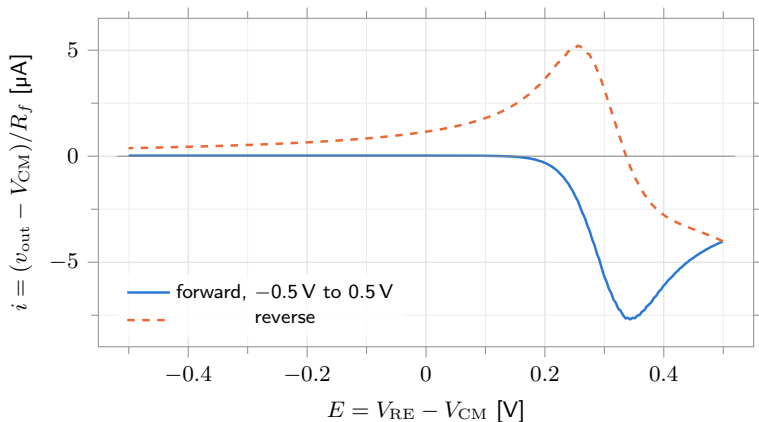
The chip

Analog front end

From currents to
a scope

References

Backup



■ schematic sim,

pre-layout

Staircase potential from the on-chip DAC, current digitised by the on-chip SAR, unclipped over the sweep.

Programmable feedback resistor: one string, 320 codes

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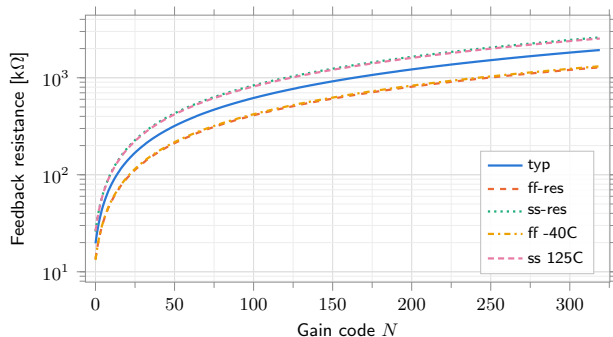
The chip

Analog front end

From currents to a score

References

Backup



$$R_f = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N$$

N	R_f	Full scale
0	19.6 kΩ	$\pm 40.7 \mu\text{A}$
63	397 kΩ	$\pm 2.02 \mu\text{A}$
319	1.93 MΩ	$\pm 0.41 \mu\text{A}$
LSB at 319		0.824 nA

Strictly monotonic at every corner, including the four thermometer seams. Maximum gain is noise-limited, not quantisation-limited.

10-bit SAR ADC, extracted netlist

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Target
application

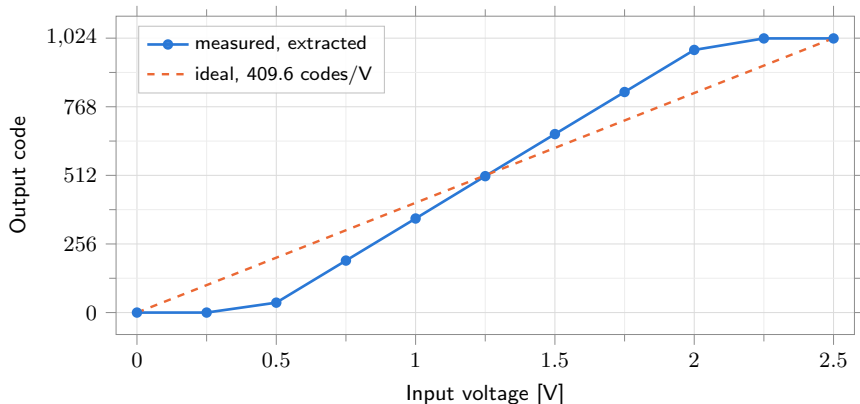
The chip

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From currents to
a scope

References

Backup



■ extracted

netlist

23 clocks per conversion, 870 kSa/s, 50 pJ. The capacitor array exists only in layout, so this is the one block characterised extracted.

Impedance spectroscopy on the same channel

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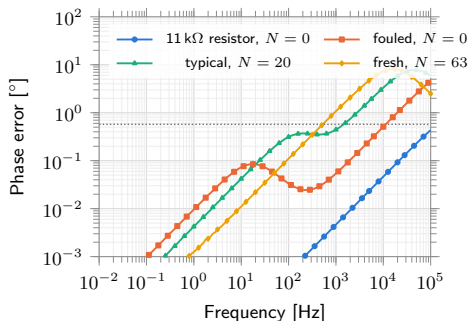
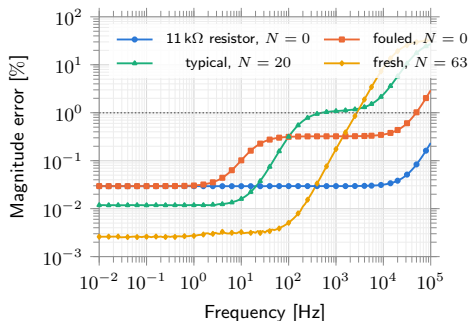
The chip

Analog front end

From currents to a scope

References

Backup



Excitation	Demodulation	Accuracy	$ Z $ range
2-code DAC square	firmware, channel hart	$\pm 1\%$, $\pm 0.6^\circ$ to 5.05 kHz; $\pm 5\%$ to 26.5 kHz	246 Ω – 12.1 M Ω

■ schematic sim, pre-layout

No EIS engine. Impedance is a ratio of two readings through one converter; the DAC drops out of the accuracy budget.

One AFE channel, as simulated

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Analog front end

From currents to a scalar

References

Backup

Parameter	Value	Condition
CE compliance	13.2 mV – 2.480 V	2.5 V rail, corners
Cell current delivered	$\pm 52.5 \mu\text{A}$	12 k Ω cell
TIA compliance	$\pm 0.80 \text{ V}$ about v_{cm}	
Feedback resistor	19.6 k Ω – 1.93 M Ω , 320 codes	monotonic at every corner
Full-scale current	$\pm 40.7 \mu\text{A}$ ($N=0$) to $\pm 0.41 \mu\text{A}$ ($N=319$)	SAR window $\cap$ TIA compliance
Current quantum	0.824 nA at maximum gain	below the 2 nA noise budget
Bipolar accuracy	0.25–0.38 LSB worst	560 k Ω , 1 M Ω electrode, corners
Control-loop PM	79–80°	five corners, into a cell
TIA-loop PM	89–93°	12-cell electrode grid
Input current noise	341 pA standalone, 3.6–21 nA into an electrode	0.1–100 Hz, 560 k Ω
Zero-current offset	$\sigma(v_{\text{re}}) = 5.72 \text{ mV}$, gain-independent	200 MC samples, calibratable
Quiescent current	197–250 μA per channel	amplifiers + local bias

■ schematic sim, pre-layout ■ Monte Carlo, 200 samples

Pre-layout, electrode held nominal: no row includes electrode-to-electrode spread.

The fusion runs on the NPU

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$$s = \sigma \left(\sum_{k=1}^4 w_k \hat{c}_k + b \right)$$

Input	peak current, peak potential, baseline $\times 4$
Mode	0, dense layer, one sigmoid
Output	one risk score
Weights	trained off-chip, staged by firmware

Per site

Hart	1, ownership enforced in hardware
Potential	own 12-bit DAC
Converter	own 10-bit SAR
Power	own MTCMOS domain

The logistic form of the cleared two-marker algorithm, extended to four inputs and moved from the central laboratory onto the electrode.

References

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Target
application

The dip

Analys front end

Front currently to
score

References

Backup

- [1] National Cancer Institute, *SEER Cancer Stat Facts*, per-site pages, seer.cancer.gov/statfacts, accessed 20 Sep. 2026. Stage and survival: SEER 21, 2016–2022; cases and deaths: 2026 estimates.
- [2] D. C. Grossman *et al.*, “Screening for ovarian cancer: USPSTF recommendation statement,” *JAMA*, vol. 319, pp. 588–594, 2018.
- [3] US Preventive Services Task Force, *A and B Recommendations; Prostate Cancer: Screening*, 2018; *Pancreatic Cancer: Screening*, 2019. uspreventiveservicestaskforce.org.
- [4] U. Menon *et al.*, “Ovarian cancer population screening and mortality after long-term follow-up in the UK Collaborative Trial of Ovarian Cancer Screening (UKCTOCS),” *Lancet*, vol. 397, pp. 2182–2193, 2021.
- [5] F. Chen *et al.*, “Clinical analysis of four serum tumor markers in 458 patients with ovarian tumors: combined use of HE4, CA125, CA19-9, and CEA,” *Cancer Manag. Res.*, vol. 10, pp. 1313–1318, 2018.
- [6] R. G. Moore *et al.*, “A novel multiple marker bioassay utilizing HE4 and CA125 for the prediction of ovarian cancer in patients with a pelvic mass,” *Gynecol. Oncol.*, vol. 112, pp. 40–46, 2009.
- [7] US Preventive Services Task Force, *Grade Definitions* (current, after July 2012), uspreventiveservicestaskforce.org, accessed 20 Sep. 2026.

Backup: USPSTF screening grades

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Grade	Certainty of net benefit	Definition [7]	Cancer screening [2, 3]
A	high, substantial	Recommends the service. Offer or provide it.	cervix (2018), colorectal 50–75 (2021)
B	high moderate, or moderate	Recommends the service. Offer or provide it.	breast 40–74 (2024), lung LDCT 50–80 (2021), colorectal 45–49 (2021)
C	at least moderate, small	Offer selectively, by professional judgment and patient preference.	prostate PSA 55–69 (2018)
D	no net benefit, or harms outweigh	Recommends <i>against</i> the service. Discourage its use.	ovary (2018) , pancreas (2019)
I	cannot be determined	Evidence insufficient, poor, or conflicting.	—
none	—	No USPSTF statement exists for the condition.	uterus

Ovarian D is a verdict on a test that was tried, CA-125 plus transvaginal ultrasound [2, 4], not on the absence of one.